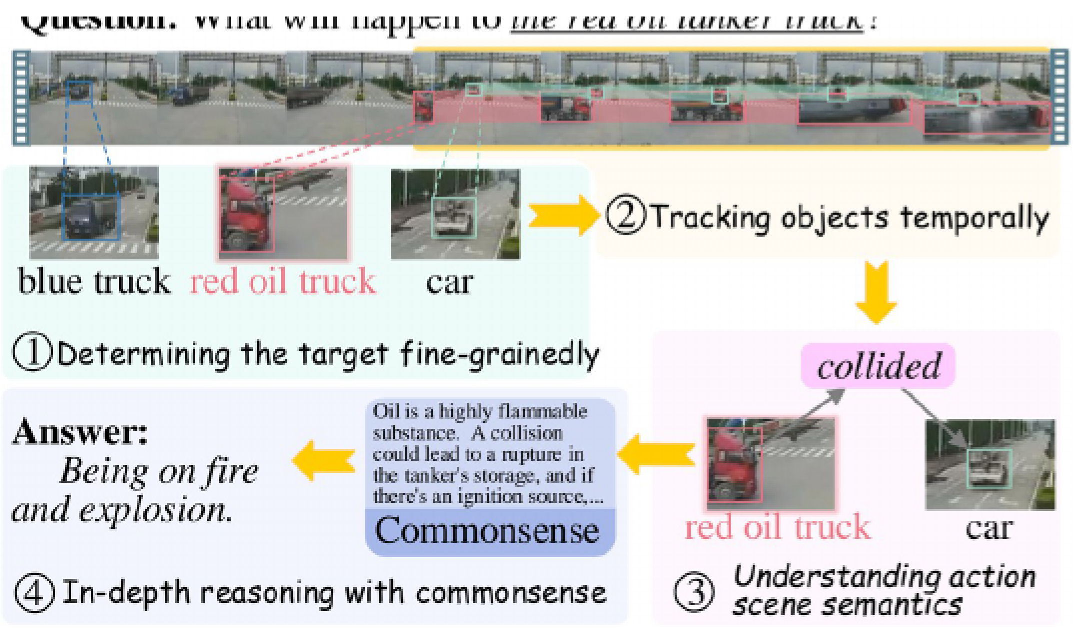
Video-of-Thought: Step-by-Step Video Reasoning from Perception to Cognition

Hao Fei; Shengqiong Wu; Wei Ji; Hanwang Zhang; Meishan Zhang; Mong Li Lee; Wynne Hsu

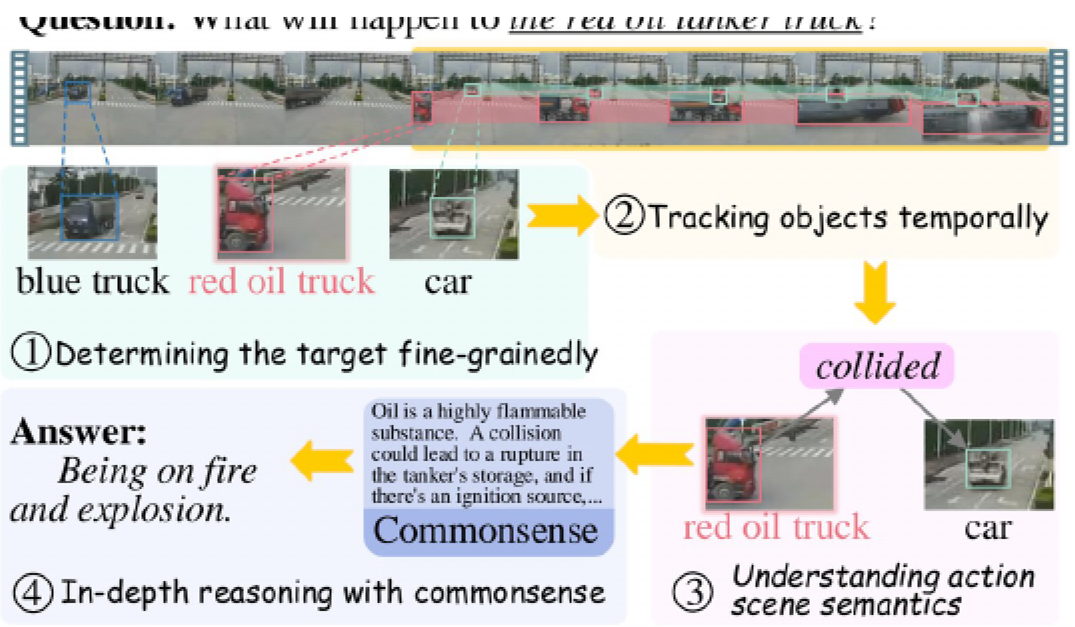
01 Motivation: From Perception to Complex Video Reasoning



- **Gap**: models excel at simple perception but struggle with complex video cognition
- Complex reasoning needs spatiotemporal understanding plus implied semantics
- Real-world applications demand reasoning over interactions, causality, and intent
- Shallow recognition is insufficient; deeper scene semantics and commonsense are required
- Challenge: bridging pixel-level cues with high-level narrative understanding

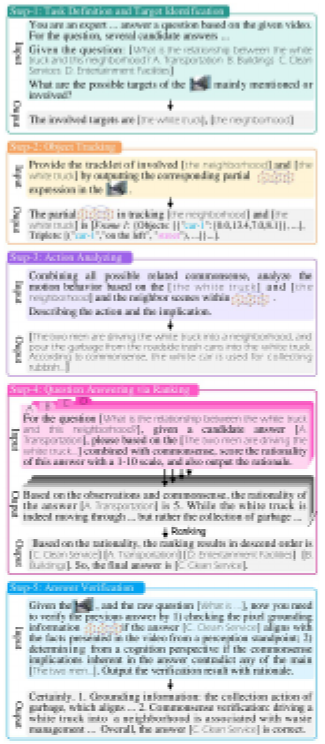
[[illegible] et al., 2018]
[Lin et al., 2019; Caballero et al., 2017]

02 Human-like Multi-step Reasoning: Core Idea



- Humans: identify → track and ground → analyze interactions → integrate commonsense
- Progression from pixel grounding to cognitive semantics with interpretable steps
- **Outcome**: coherent, verified answers with explicit rationales
- Benefit: reduces ambiguity by decomposing complex queries
- Targets, trajectories, and relations form the backbone of reasoning

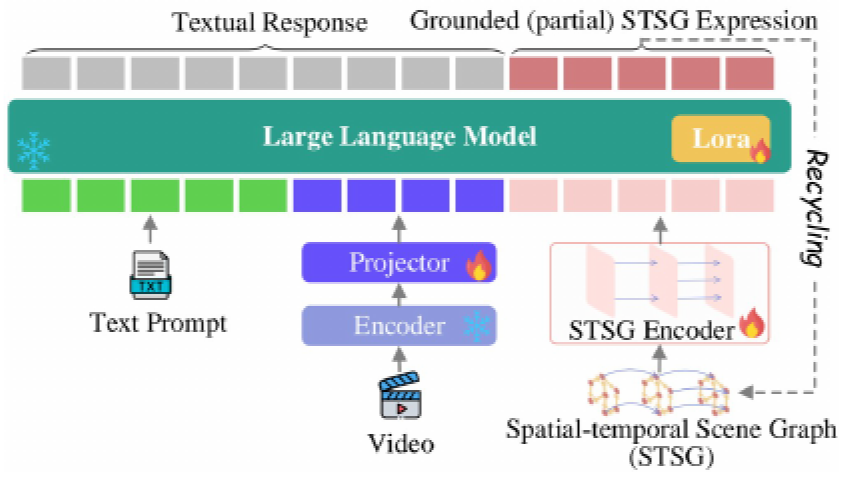
03 CoT for Video MLLMs is Under-explored



- MLLMs advanced; video MLLMs emerging
- Classic CoT helps text tasks; video-specific CoT scarce
- Need a video-native CoT that decomposes problems with temporal grounding
- Limitation: naive step-by-step prompts underperform for video
- **Opportunity**: combine structure from scene graphs with LLM commonsense

[Zhang et al., 2023a; Maaz et al., 2023; Lin et al., 2023]
[Wei et al., 2022; Wang et al., 2022a]

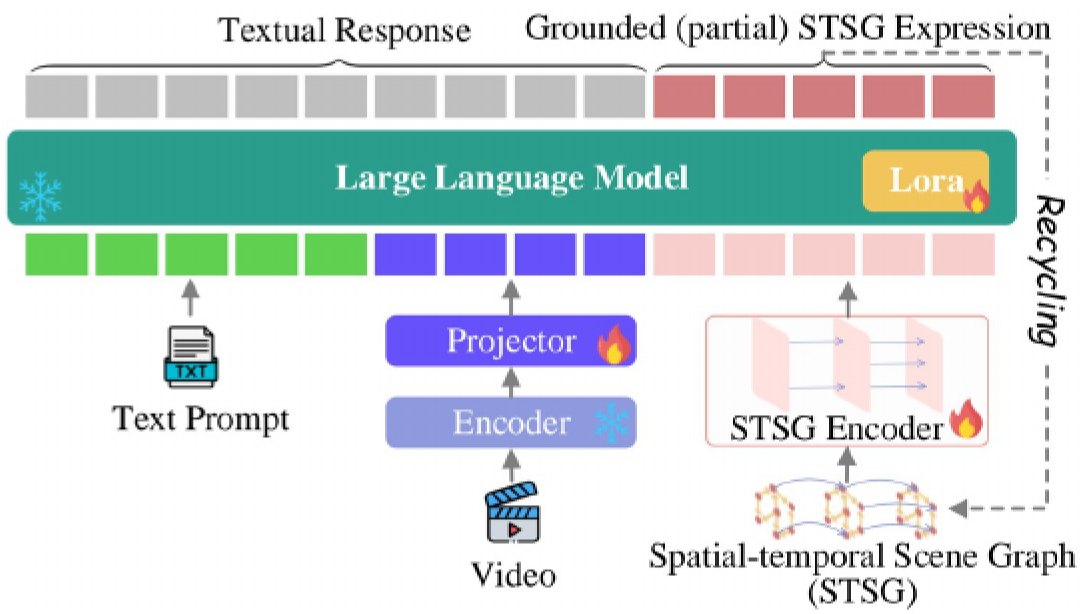
04 MotionEpic + Video-of-Thought (VoT)



- **MotionEpic**: video MLLM that understands and generates STSGs
- **VoT**: chains video reasoning from grounding to cognition
- Grounding-aware tuning enables STSG-free downstream inference
- Gain: interpretable rationales through generated structure
- Unified pipeline for perception, interaction analysis, and answer verification

[Ji et al., 2020]

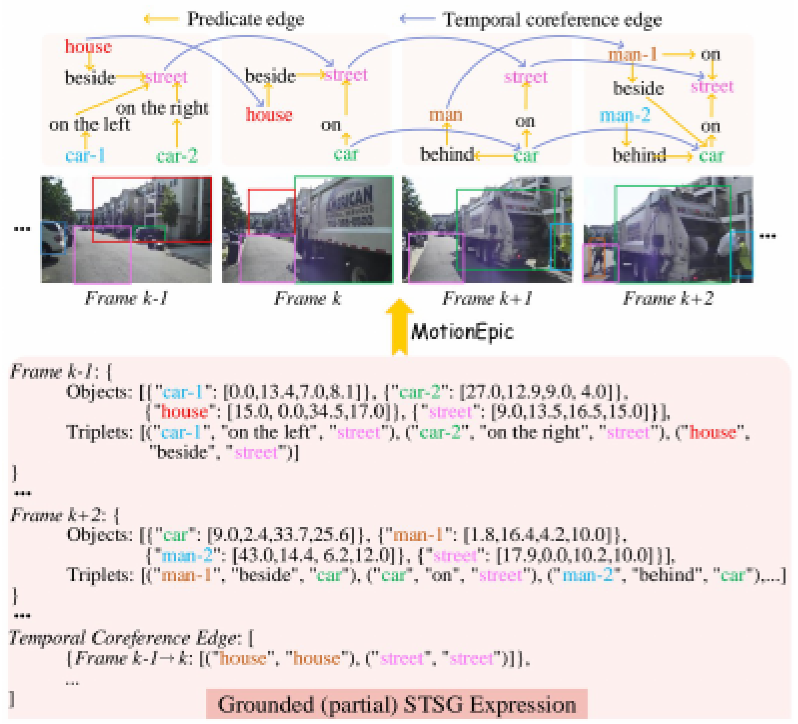
05 MotionEpic Overview and Inputs



- Inputs: text, video, and STSG
- Backbone: Vicuna-7B; Video: ViT-L/14 plus Q-Former; STSG: recurrent Graph Transformer
- Aligned multi-source encodings for grounding-aware reasoning
- **Design**: structured nodes and relations augment dense visual tokens
- Effect: improved alignment between content and semantics

[Chiang et al., 2023; Dosovitskiy et al., 2020; Li et al., 2023a]
[Dwivedi & Bresson, 2020]

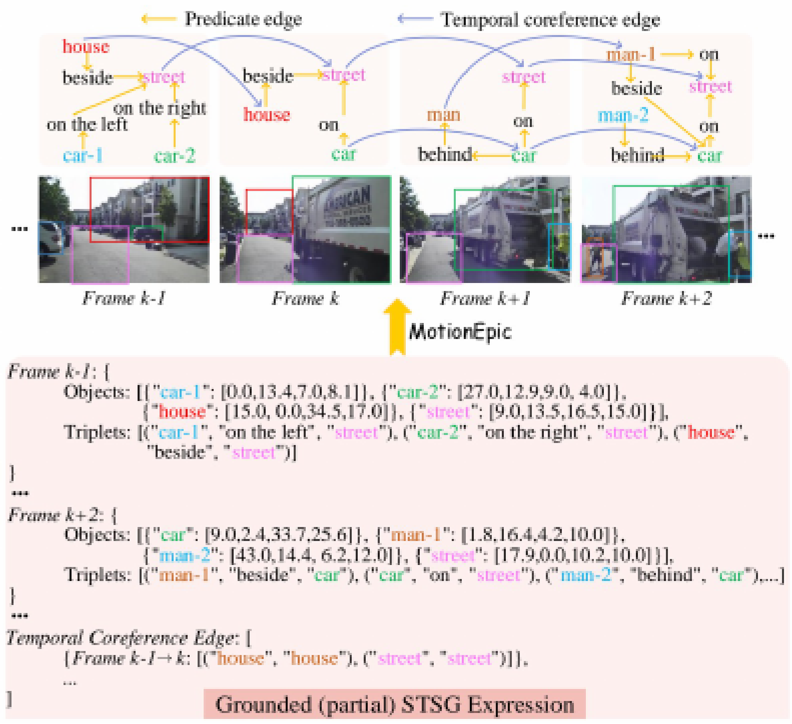
06 Representation and Encoding Design



- STSG: per-frame scene graphs plus temporal coreference edges
- Nodes carry category, features, and bounding box; edges capture predicates and tracking
- Sampling and recurrent propagation improve efficiency and temporal continuity
- **Goal**: robust trajectories and relations for long-range reasoning
- Risk: over-sampling adds redundancy and noise; balanced rate preferred

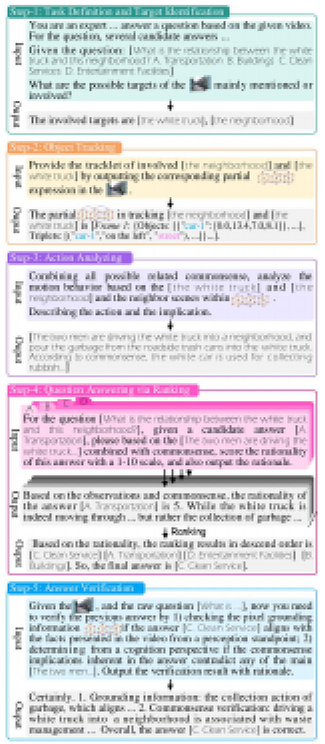
[Ji et al., 2020]

07 Grounded Generation and STSG-free Inference



- Model can generate partial STSG as rationale
- Grounding-aware tuning enables STSG-free downstream inference
- Coarse-to-fine objectives align video and STSG for accurate parsing
- **Recycle**: generated structure feeds subsequent reasoning rounds
- Outcome: factual grounding reduces hallucination in answers

08 Video-of-Thought: Perception-to-Cognition Chain



- Steps: target select → tracklet grounding → interaction analysis → candidate ranking → dual verification
- Verification: pixel grounding plus commonsense
- Decomposition yields factual, explainable answers for complex VideoQA
- **Key idea**: solve low-level perception before high-level reasoning
- Impact: stronger robustness on long and cluttered videos

[Wei et al., 2022]

09 Main Results Overview and Key Observations

Model	VLEP	STAR				IntentQA	Social-IQ	
		Int.	Seq.	Pre.	Fea.		2-Way	4-Way
• **SoTA baselines**								
InternVideo	63.9	62.7	65.6	54.9	51.9	-	-	-
LLaMA-VQA	71.0	66.2	67.9	57.2	52.7	-	-	-
VLAP	69.6	70.0	70.4	65.9	62.2	-	-	-
SeViLA	68.9	63.7	70.4	63.1	62.4	-	-	-
VideoChat	62.0	63.2	66.8	54.1	49.6	59.3	67.7	37.8
Video-LLaVA	65.8	64.3	67.0	56.5	50.1	62.5	68.9	39.2
• **CoT**								
Video-LLaVA	65.7	65.0	67.7	57.8	52.0	63.2	69.5	40.4
Video-LLaVA+STSG	67.0	65.9	68.9	58.7	53.7	64.9	70.4	41.7
MotionEpic	68.2	66.8	69.6	60.6	57.4	66.1	71.7	43.0
• **VoT**								
MotionEpic	**73.4**	**71.5**	**72.6**	**66.6**	**62.7**	**70.8**	**72.8**	**45.0**

- **Overall**: VoT plus MotionEpic outperforms state of the art across complex VideoQA
- STSG integration and VoT surpass vanilla CoT approaches
- Implicit STSG integration in MotionEpic beats explicit feature fusion
- Consistent gains: fine-tuning and zero-shot settings
- CoT alone shows limited lift for video reasoning compared to VoT

10 Dataset-specific: Causal-VidQA and NExT-QA

Model	Acc@D	Acc@E	Acc@P			Acc@C		
			A	R	AR	A	R	AR
• **SoTA baselines**								
TranSTR	73.6	75.8	65.1	65.0	48.9	68.6	65.3	50.3
Video-LLaMA	69.2	71.0	63.6	62.4	44.4	65.4	60.1	45.0
VideoChat	72.9	73.9	65.2	63.1	45.9	66.0	62.7	45.8
Video-ChatGPT	73.1	75.1	66.0	63.9	46.0	67.8	63.6	50.0
Video-LLaVA	73.7	74.4	67.6	65.4	47.7	68.0	64.9	51.5
• **CoT**								
Video-LLaVA	74.2	74.8	68.0	65.7	48.1	70.3	65.7	52.9
Video-LLaVA+STSG	75.7	75.9	68.9	67.2	50.0	70.7	67.2	53.6
MotionEpic	78.5	77.2	70.1	70.8	52.4	71.2	69.1	55.0
• **VoT**								
MotionEpic	**81.2**	**83.0**	**74.3**	**73.7**	**54.7**	**74.5**	**73.8**	**58.6**

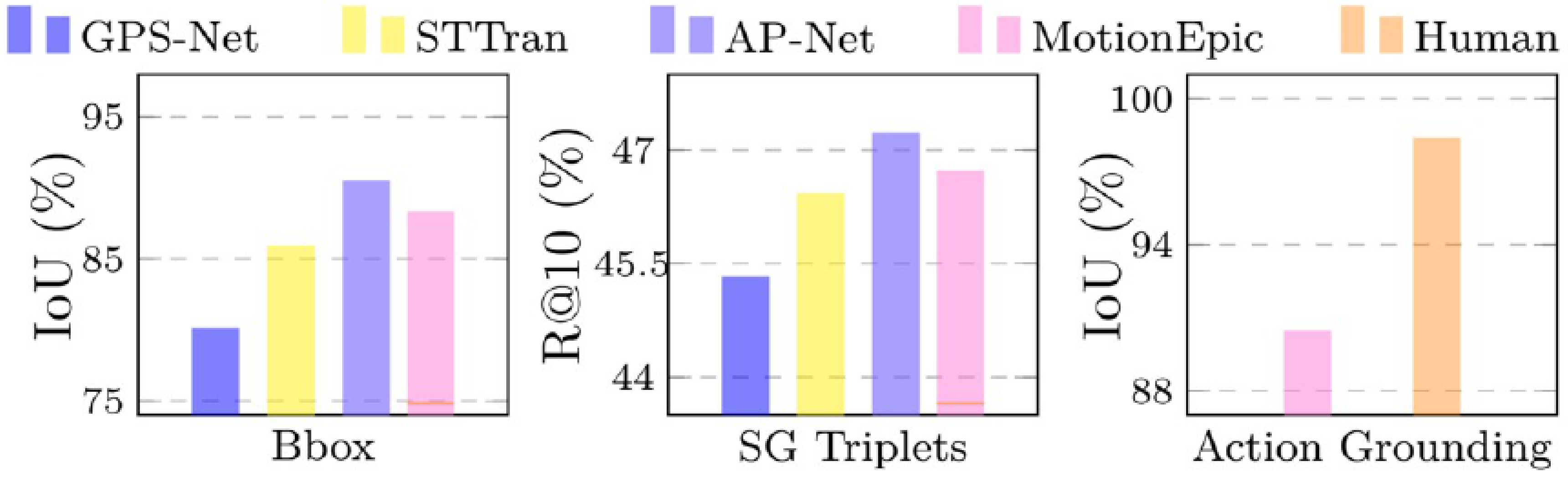
- Large margins on causal and complex reasoning subsets
- Benefits in description, explanation, prediction, and counterfactual
- Consistent gains across Acc@All, causal, temporal, and descriptive splits
- Stronger: reasoning about interactions and intentions
- Stable improvements across multiple baselines and settings

11 Zero-shot Performance and Verification Ablations

Model	MSR-VTT	ActivityNet	NExT-QA	STAR	*AVG.*
• **Zero-shot SoTA baselines**					
InternVideo	-	-	49.1	41.6	-
Video-LLaMA	49.6	21.4	43.5	36.4	37.7
VideoChat	52.0	26.5	52.8	45.0	44.1
Video-ChatGPT	54.3	35.2	53.0	48.7	47.8
Video-LLaVA	59.2	45.3	57.3	50.6	53.1
VideoChat2	54.1	49.1	61.7	59.0	56.0
• **CoT**					
Video-LLaVA	60.0	46.9	59.5	52.0	54.6
Video-LLaVA+STSG	61.5	48.4	60.6	52.7	55.8
MotionEpic	63.1	50.0	61.9	56.5	57.8
• **VoT**					
MotionEpic	**66.2**	**54.6**	**66.5**	**61.7**	**62.3**
w/o Verify-G	63.6	51.4	62.0	59.1	59.0
w/o Verify-C	65.1	53.4	62.8	58.8	60.1

- VoT beats CoT in zero-shot; verification boosts robustness
- Grounding verification critical for perceptive datasets; both matter for complex videos
- Dual verification yields clearest gains on complex QA (NExT-QA, STAR)
- **Finding**: VoT benefits grow when no in-domain training is used
- Net effect: higher accuracy through evidence-backed answers

12 Grounding Probes and Ablations



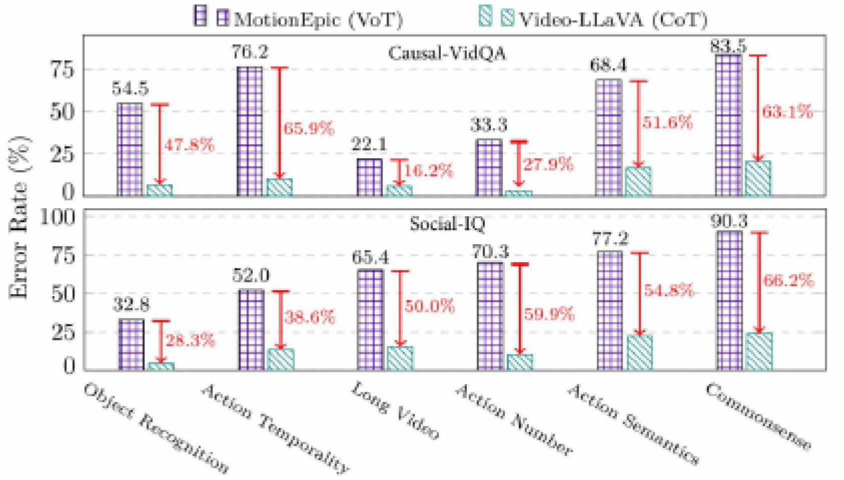
- MotionEpic competitive with STSG parsers on grounding tasks
- Ablations reveal which grounding objectives matter most
- Reliable grounding supports downstream multi-step reasoning
- Evidence: strong object, relation, and action alignment
- **Takeaway**: grounding quality correlates with reasoning accuracy

[Lin et al., 2020; Cong et al., 2021; Li et al., 2022b]

13 Error Breakdown and Human Eval

- MotionEpic plus VoT approaches human performance on hard subsets
- Error reductions in action semantics and commonsense
- Human study validates VoT gains over CoT baselines
- Improved: interpretability and answer faithfulness
- **Focus**: fewer mistakes in multi-entity interactions

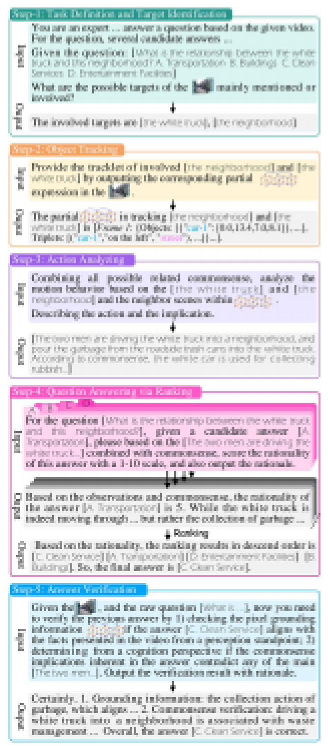
Data	CoT		VoT	Human
	Video-LLaVA	MotionEpic	MotionEpic	
Causal-VidQA	32.4	56.8	74.3	80.6
Social-IQ	22.3	40.1	61.4	72.7



14 End-to-end VoT Case

- STSG grounding reduces hallucination
- Sequential reasoning and verification yields correct answer
- Explainable chain from perception to cognition
- **Rationale**: grounded entities, relations, and actions
- Clarity: traceable steps support human auditing

15 VoT + MotionEpic—From Grounding to Cognition



- **VoT**: first video-native CoT framework with dual verification
- **MotionEpic**: STSG-grounded video MLLM enabling STSG-free inference
- State-of-the-art across complex VideoQA; broad implications
- Outlook: transferable to richer multimodal reasoning tasks
- Bridging perception and cognition for reliable video understanding

THANKS!